Project Proposal: Data Engineering Pipeline

Caroline Clark

5/17/2026

1. Project Title

Weather Data Engineering Pipeline: API Extraction, Transformation, and Interactive Visualization

2. Introduction and Background

- Weather data plays a large role in planning, safety, transportation, environmental monitoring and much more. However, raw weather information from APIs is often difficult to analyze, and not easily accessible to the average user. This project's goal is to build a complete, end to end data engineering pipeline that automatically accesses weather data from a public API, processes it into a clean dataset, and visualizes it through an interactive dashboard. This topic is important because weather patterns influence daily personal and business decisions. Also, this helps gain accessible data for students or anyone wanting to learn about, or research environmental data.

3. Problem Statement

- Although weather APIs provide in depth environmental data, the information is delivered in raw JSON formats that are not ready for analysis. Users need a simple, automated system that gathers weather data reliably, cleans and structures it and presents it in a simple, interactive dashboard. This project creates a useful, beginner friendly pipeline that turns raw API data into meaningful insights.

4. Objectives

The project will…

- Get hourly and daily weather data from the Open-Meteo API
- Transform JSON responses into structured tables using Python and pandas
- Clean the data by handling missing values and converting timestamps
- Load the processed data into CSV files
- Visualize weather trends using Dash or Power BI

- Enable interactivity such as date filtering and metric selection
- Create insights on temperature, precipitation, and wind patterns

5. Methodology / Technical Approach

API / Data Source

- Open-Meteo API- free, stable, no authentication required

Tools & Technologies

- Python (Libraries: request, pandas, json, matplotlib or plotly)
- Storage: CSV or SQLite
- Visualization: Dash or Microsoft Power BI
- Version control: GitHub

ETL Workflow

Extract

- Use Python's requests library to access the Api
- Get hourly and daily data such as temperature, precipitation, humidity, and wind speed

Transform

- Convert JSON into pandas DataFrames
- Clean missing or inconsistent values
- Convert timestamps
- Create fields such as daily averages, weather categories, etc.

Load

- Save cleaned datasets into CSV files or load them into SQLite
- Maintain a historical dataset for trend analysis

Visualize

- Build an interactive dashboard with temperature trend line chart, daily precipitation bar chart, wind speed visualization, high/low temperature, total rainfall, date range filters

ETL Architecture Diagram

❖ Open-Meteo API

- Python Extraction Script
  - Data Cleaning and Transformation by pandas
    - Storage in CSV or SQLite
      - Interactive Dashboard in Dash or Power BI

6. Timeline

Week 1	:Select API, explore use of APIs, create proposal
Week 2	: Build extraction script, analyze JSON responses
Week 3:	Data cleaning and storage layer
Week 4:	Create dashboard and visualizations
Week 5	:Final testing, prepare presentation

7. Expected Outcomes

By the end of the project, we will have…

- A functioning ETL pipeline that automatically retrieves and processes weather data
- Clean, structured datasets stored in CSV or SQLite
- An interactive dashboard with multiple weather visualizations
- A presentation explaining the workflow, challenges, and insights